🫁 LungPair: Preclinical Drug-Pair Triage

Multi-Agent Gemini Architecture with Calibrated Refusal Boundaries

AgentVerse 2026 · Biotech Track
Team: *Backprop to the Future* · Slide 42 Deliverable
Code: **github.com/leventkoymen/backprop-to-the-future-...**

1. THE PRECLINICAL BOTTLENECK — 3.3% BASE RATE

In metastatic non-small cell lung cancer (NSCLC), single-agent targeted therapies (e.g. osimertinib) inevitably fail due to acquired bypass resistance. Combination doublets are the biological imperative, but combinations create an impossible combinatorial explosion: 105 panel drugs generate **5,460 potential pairs**.

In the landmark NCI-ALMANAC matrix across 9 lung cancer cell lines, researchers screened 5,228 pairs (47,052 robotic experiments). **Only 172 pairs (3.3%) demonstrated validated synergy** (ComboScore > 50). Over 96% were dead ends. At $340 per experiment ($3,060/pair), blind wet-lab screening costs **$16.0M**. Preclinical labs lack calibrated computational triage that prioritizes hits while reliably refusing when evidence is absent.

2. INTENDED USERS & COMMERCIAL MODEL — STRICTLY B2B

Primary Users: Academic oncology screening cores, translational cancer centers (Crick, ICR, NCI), and preclinical CROs (Charles River, WuXi).

Commercial Model & Procurement Velocity: B2B SaaS recharge licence ($35,000/core/yr). Sells directly out of departmental research consumable budgets in 4–8 weeks, completely bypassing 3-year hospital clinical procurement drag. A single 50-pair triage campaign saves **$147,000 in prevented consumable waste** ($93,000 down to $4,900/hit), delivering an immediate 4.2x ROI.

3. PERMITTED DATA, APIS & GOVERNANCE — AUDIT IN DATA.MD

- **NCI-ALMANAC (DTP):** 5,228 measured pairs across 9 NSCLC lines (A549, EKVX, HOP-62, HOP-92, H226, H23, H322M, H460, H522). Public Domain.
- **PubChem PUG-REST API:** Real-time conversion of drug names to canonical SMILES, CID, molecular formulas, and 3D SDF conformers. Public Domain.
- **EMBL-EBI ChEMBL API:** REST retrieval of validated biological targets (e.g. EGFR, TUBB1) to prime downstream literature grounding. CC BY-SA 4.0.
- **NCBI PubMed E-Utilities:** Dynamic mechanistic literature retrieval. Public Domain.
- **Zero Patient Data:** 100% public *in vitro* cell-line screens. Strict RUO boundary.

4. LIVE THREE.JS WEB PLATFORM — INTERACTIVE 3D DEMO

Hosted live at: **yonghwa11701-hash.github.io/lungpair-3d-visual/**. Standalone 2.4 MB client tool running in any browser across 3 synchronized scenes:

Scene 1 (Haystack): Spatial 3D matrix of all 5,228 pairs; hit key lights up 172 hits.
Scene 2 (Agent Triage): Multi-agent proposal cards, PubChem 3D stick models, and mandatory human sign-off before batch sealing.
Scene 3 (Dose Surface): 3D dose-response surface mapped over the Bliss independence manifold to confirm clinical plasma ($C_{\max}$) reachability.

5. MULTI-AGENT ENGINE — GOOGLE DEEPMIND HARNESS²

Built under Google DeepMind's Harness² paradigm (arXiv:2610.14224)—enforcing external deterministic gates over language models:

- **Orchestrator (`gemini-4-argon`):** Dispatches PubChem entity resolution and primes targets via ChEMBL.
- **Offline Random Forest Ranker:** Predicts synergy scores and 95% intervals from frozen chemical-transcriptomic matrices. *The LLM never calculates numbers.*
- **Research Pool (`gemini-3.8-flash`):** Parallel literature grounding threads under a 90s budget searching PubMed for mechanistic synergy.
- **Checker (`gemini-3.5-flash-lite`):** Audits outputs against 8 Python gates.

6. HELD-OUT EVALUATION & CALIBRATED TIERS — 51% NOISE CAP

Evidence Tier	Clinical Definition	Precision	Lift
Tier 1	Pair tested in another cancer	**0.89**	**24.0x**
Tier 2	Both drugs known, pair untested	**0.34**	**11.0x**
Tier 3	Monotherapy data only	0.06	3.0x
Tier 4	Unseen chemical space	0.04	**REFUSED**

Held-Out Cross-Validation (5 Seeds): Precision@20 = **0.89**, Precision@50 = **0.61**, AUPRC = **0.82**, MCC@50 = **0.71** against 0.028 random baseline.
Assay Noise Cap: Biological replicate agreement in NCI-ALMANAC is ~51%; our held-out performance reaches this ceiling without overfitting.

Published Negative Results (Scientific Honesty):
1. *Genotype Matching:* KRAS/TP53 mutation matching showed no signal over random same-size cell lines (win rate 28–59%).
2. *Merck External Benchmark (132 pairs):* NCI direct lookup beat model (50% vs 40%), proving bounded transfer. Tier 4 is refused for this reason.

7. 8 DETERMINISTIC PYTHON CODE GATES — SAFETY SYSTEM

- **1. Pre-Model Intent Gate:** Regex intercepts dosing queries before LLM call.
- **2. Biologics Refusal:** Deterministically blocks antibodies (pembrolizumab).
- **3. Grounded Citations Gate:** Strips any PMID not verified in search redirects.
- **4. Exact-Match Numbers Gate:** Natural language decimals match ranker bit-for-bit.
- **5. Calibrated Tier 4 Refusal:** Unseen pairs withheld: *"Test monotherapy first."*
- **6. Prompt Sanitization:** Blocks prompt extraction and injection overrides.
- **7. Human Sign-Off Gate:** Halts at terminal; requires explicit researcher approval.
- **8. Offline Replay Engine:** Zero-latency execution (`--replay`) with 25 unit tests.

Regulatory & Health Economics: Research Use Only (RUO) now; future serial ctDNA liquid biopsy enters MHRA/FDA SaMD. Compresses screening cost from $93k to $4,900/hit, aligned with NICE's £30k/QALY cost-effectiveness ceiling.

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Terminal: `./demo.sh 1` · Replay: `./demo.sh --replay` · 25/25 Tests Pass

Web App: **yonghwa11701-hash.github.io/lungpair-3d-visual/**